Junayed Mahmud

L3Harris Engineering Center, HEC 249, Orlando, FL, 32816
junayed.mahmud@ucf.edu +1 (571) 595-2727 jmahmud47.github.io
Google scholar www.linkedin.com/in/junayed-mahmud/ Green card holder

TECHNICAL SKILLS

- Programming Languages: Python, Java, C, C++, Swift, Objective C, Perl, Shell Scripting, Kotlin, JavaScript, R, MATLAB, PHP, HTML
- Machine Learning/Deep Learning Frameworks: LangChain, Pytorch, Tensorflow, HuggingFace
- Python Scientific Stack: Numpy, Pandas, Scipy, Sklearn, Jupyter
- CI/CD & DevOps Tools: Git, Jira, Jenkins, Docker, Kubernetes, GitLab
- Software Development Frameworks and Tools: Flask, Probot, Visual Studio Code, Android Studio, Xcode, Django, Spring, AngularJS, Hibernate, Bootstrap, NodeJS, Unity
- Databases/SQL & Storage: Oracle, MySQL, PostgreSQL, MongoDB, SQLite, NoSQL

SELECTED PROJECTS

Utilizing GUIs for Identifying Android Bugs in Code Paper Code Dataset
This project aims to identify bugs in code for Android apps utilizing information from graphical user interfaces (GUIs). We adapted two text-retrieval (TR)-based and two neural text embedding approaches. Our study outperformed existing approaches by a marked increase in Hits@10 of 13-18%. Later, we develop a Github bot using the Probot framework and a Flask back-end to automate bug localization.

Assessing the Quality of Bug Reproduction Steps Paper Code
This project assesses the quality of the bug reproduction steps on Github by mapping information to GUIs. For analyzing bug reports and mapping steps to GUIs, we integrated GPT-4 using langchain framework. Our approach annotates bug reproduction steps 25.2% better (in terms of F1 score) than the state-of-the-art.

Evaluating Language Models for Code Paper Code
This project automatically generated comments from Java methods using language models. We quantitatively and qualitatively evaluated the limitations of existing machine translation metrics and proposed a taxonomy of errors.

RESEARCH INTERESTS

Software Engineering, Bug Reporting, Bug Localization, Program Repair, Automated Mobile Testing, Natural Language Processing for Software Engineering, Source Code Analysis

EDUCATION

Ph.D. in Computer Science Aug 2019 – Present
University of Central Florida, Florida, USA
- Advisor: Dr. Kevin Moran
- George Mason University (Aug 19 – Aug 23) and University of Central Florida (Aug 23 – Present)
- Dissertation: Multimodal Learning for Automated Bug Report Management
- CGPA: 4.00 (out of 4.00)

M.S. in Computer Science May 2023
George Mason University, Virginia, USA
- Advisor: Dr. Kevin Moran
- CGPA: 3.98 (out of 4.00)

B.S. in Computer Science and Engineering Nov 2016
Islamic University of Technology, Dhaka, Bangladesh
- Advisor: Dr. Abu Raihan Mostofa Kamal
- CGPA: 3.49 (out of 4.00)

WORK EXPERIENCE

University of Central Florida, Florida, USA
- *Graduate Research Assistant* Aug 2023 – Present
  - Leveraged large language models (LLMs) and agentic AI for program repair on our new dataset of 71 Android graphical user interface (GUI) bugs and automated the testing of the generated bug fixes
  - Automatically evaluated 75 annotated bug reports by mapping them to the GUI screens using LLMs via LangChain framework, prompt engineering, and program analysis to provide feedback to reporters

- Utilized LLMs to automatically generate unit and UI tests to validate the existence of diverse types of reported bugs across 24 Android applications to aid in regression testing
- Addressed the limitations of three code-to-comment generative AI models and improved the quality of software documentation using transformer architecture and contrastive learning

George Mason University, Virginia, USA

- *Graduate Research/Teaching Assistant* Aug 2019 – Aug 2023
  - Adapted information-retrieval and neural text-embedding approaches for GUI bug localization on our new dataset of 80 bugs from 39 apps to automate debugging, with the pipeline containerized via docker
  - Developed a dynamic program analysis tool that converts user-performed app actions into replayable scenarios and extracts GUI data from 51 Android applications for automated testing
  - Built a chatbot for bug reporting and conducted a user study to understand the usefulness of the tool in improving bug report quality among 36 participants using 12 Android applications
  - Studied the characteristics of diverse types of 180 reproducible bug reports to build effective automated techniques for different bug report management activities
  - Fine-tuned image captioning models on 10204 GUI screens, using visual software data encoded in GUIs to understand the usefulness of GUI-centric software documentation
  - Fine-tuned code-to-comment translation models on 500000 function-comment pairs to understand the limitations in existing models and proposed an error taxonomy by performing a qualitative study
  - Assisted two courses on Systems Programming and C++

Samsung R&D Institute Bangladesh Ltd., Dhaka, Bangladesh

- *Software Engineer* Jan 2017 – Mar 2019
  - Worked in a team following agile methodology for developing, testing, and releasing an iOS application named SmartThings, which 285M+ users use to control electronic devices via smartphones
  - Developed a backend architecture named IoTivity, which is based on REST API and enables device communication within the Internet of Things (IoT) system
  - Developed multiple frontend GUIs for the SmartThings project

REFERRED PUBLICATIONS

15. **[Under Review] J. Mahmud**, S. Pandey, N. D. Silva, A. K. Dipongkor, O. Chaparro, M. Fazzini, and K. Moran, *"Automated Program Repair for UI-centric Android Bugs: How Far are We?,"* 22 pages.
14. [ASE'25] J. Johnson, **J. Mahmud**, O. Chaparro, K. Moran, and M. Fazzini, "Generating Failure-Based Oracles to Support Testing of Reported Bugs in Android Apps," in *Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering*, Seoul, South Korea, Nov 2025, pp. 2196-2208. (***10% acceptance rate***)
13. [ICSME'25] **J. Mahmud**, J. Chen, T. Achille, C. Alvarez-Velez, D. Bansil, P. Ijieh, S. Karanch, N. D. Silva, O. Chaparro, A. Marcus, and K. Moran, "LadyBug: A GitHub Bot for UI-Enhanced Bug Localization in Mobile Apps," in *Proceedings of the IEEE 41st International Conference on Software Maintenance and Evolution*, Tool Demonstration Track, Auckland, New Zealand, Sep 2025, pp. 1-5.
12. [ICPC'25] **J. Mahmud**, A. Saha, O. Chaparro, K. Moran, and A. Marcus, "Combining Language and App UI Analysis for the Automated Assessment of Bug Reproduction Steps," in *Proceedings of the 33rd IEEE/ACM International Conference on Program Comprehension*, Ottawa, Canada, Apr 2025, pp. 1-12. (***41% acceptance rate***)
11. [ISSTA'24] A. Saha, Y. Song, **J. Mahmud**, Y. Zhou, K. Moran, and O. Chaparro, "Toward the Automated Localization of Buggy Mobile App UIs from Bug Descriptions," in *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis*, Vienna, Austria, Sep 2024, pp. 1249-1261. (***21% acceptance rate***)
10. [ICSE'24] **J. Mahmud**, "Toward Rapid Bug Resolution for Android Apps," in *Proceedings of the 46th IEEE/ACM International Conference on Software Engineering*, Doctoral Symposium Track, Lisbon, Portugal, Apr 2024, pp. 237-241. (***57% acceptance rate***)
9. [ICSE'24] **J. Mahmud**, N. D. Silva, S. A. Khan, S. H. Mostafavi, S. M. H. Mansur, O. Chaparro, A. Marcus, and K. Moran, "On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization," in *Proceedings of the 46th IEEE/ACM International Conference on Software Engineering*, Lisbon, Portugal, Apr 2024, pp. 1-13. (***7% acceptance rate***)

8. [MSR'24] K. Baral, J. Johnson, **J. Mahmud**, S. Salma, M. Fazzini, J. Rubin, J. Offutt, and K. Moran, "Automating GUI-based Test Oracles for Mobile Apps," in *Proceedings of the 21st International Conference on Mining Software Repositories*, Lisbon, Portugal, Apr 2024, pp. 309-321. (***29% acceptance rate***)

7. [ICSE'23] Y. Song, **J. Mahmud**, N. D. Silva, Y. Zhou, O. Chaparro, K. Moran, A. Marcus, and D. Poshyvanyk, "BURT: A Chatbot for Interactive Bug Reporting," in *Proceedings of the 45th IEEE/ACM International Conference on Software Engineering*, Formal Tool Demonstrations Track, Melbourne, Australia, May 2023, pp. 170-174. (***48% acceptance rate***)

6. [FSE'22] Y. Song, **J. Mahmud**, Y. Zhou, O. Chaparro, K. Moran, A. Marcus, and D. Poshyvanyk, "Toward Interactive Bug Reporting for (Android App) End Users," in *Proceedings of the 2022 ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, Singapore, Nov 2022, pp. 344-356. (***22% acceptance rate***)

5. [SANER'22] J. Johnson, **J. Mahmud**, T. Wendland, K. Moran, J. Rubin and M. Fazzini, "An Empirical Investigation into the Reproduction of Bug Reports for Android Apps," in *Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering*, Honolulu, Hawaii, Mar 2022, pp. 321-332. (***24% acceptance rate***)

4. [SANER'22] K. Moran, A. Yachnes, G. Purnell, **J. Mahmud**, M. Tufano, C. B. Cardenas, D. Poshyvanyk, and Z. H'Doubler, "An Empirical Investigation into the Use of Image Captioning for Automated Software Documentation," in *Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering*, Honolulu, Hawaii, Mar 2022, pp. 514-525. (***24% acceptance rate***)

3. [ACL'21] **J. Mahmud**, F. Faisal, R. I. Arnob, A. Anastasopoulos, and K. Moran, "Code to Comment Translation: A Comparative Study on Model Effectiveness & Errors," in *Proceedings of the First Workshop on Natural Language Processing for Programming*, Co-located with ACL-IJCNLP'21, Bangkok, Thailand, Aug 2021, pp. 1-16.

2. [MSR'21] T. Wendland, J. Sun, **J. Mahmud**, S. M. H. Mansur, S. Huang, K. Moran, J. Rubin and M. Fazzini, "AndroR2: A Dataset of Manually-Reproduced Bug Reports for Android Apps," in *Proceedings of the 18th Conference on Mining Software Repositories*, Data showcase track, Madrid, Spain, May 2021, pp. 600-604.

1. [SAS'18] A. R. Chowdhury, **J. Mahmud**, A. R. M. Kamal, and M. A. Hamid, "MAES: Modified Advanced Encryption Standard for Resource Constraint Environments," in *Proceedings of the 2018 IEEE Sensors Applications Symposium*, Seoul, Korea (South), Mar 2018, pp. 1–6.

TALKS & FORMAL PRESENTATIONS

- **Research Paper Presentation** - Software Engineering Seminar, University of Central Florida, Florida, USA — Feb 10, 2025
  - "Combining Language and App UI Analysis for the Automated Assessment of Bug Reproduction Steps"
- **Dissertation Proposal Presentation** - University of Central Florida, Florida, USA — Dec 04, 2024
  - "Multimodal Learning for Automated Bug Report Management"
- **Research Paper Presentation** - Software Engineering Seminar, University of Central Florida, Florida, USA — Mar 27, 2024
  - "On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization"
- **Research Paper Presentation** - Software Engineering Class (CEN 5016), University of Central Florida, Florida, USA — Mar 7, 2024
  - "Code to Comment Translation: A Comparative Study on Model Effectiveness & Errors"
- **Comprehensive Exam Presentation** - George Mason University, Virginia, USA — Apr 29, 2022
  - "Automating Bug Report Management: A Survey"
- **Research Paper Presentation** - Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER'22), Virtual (originally Honolulu, Hawaii) — Mar 16, 2022
  - "An Empirical Investigation into the Use of Image Captioning for Automated Software Documentation"
- **Research Paper Presentation** - Software Engineering Seminar, George Mason University, Virginia, USA — Nov 11, 2021

- "Code to Comment Translation: A Comparative Study on Model Effectiveness & Errors"

- **Research Paper Presentation** - Proceedings of the First Workshop on Natural Language Processing for Programming (NLP4Prog'21), Co-located with ACL-IJCNLP'21, Virtual (originally Bangkok, Thailand) Aug 06, 2021
  - "Code to Comment Translation: A Comparative Study on Model Effectiveness & Errors"

PROFESSIONAL SERVICES

External Reviewer

- 47th IEEE/ACM International Conference on Software Engineering (ICSE'25)
- 37th IEEE/ACM International Conference on Automated Software Engineering (ASE'22)
- 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER'22)
- 29th IEEE/ACM International Conference on Program Comprehension (ICPC'21)
- 2021 Mining Software Repositories (MSR'21)

HONORS & AWARDS

- Dissertation Fellowship Award — Nov 2025
- Outstanding Graduate Creative Work Award — Jan 2024
- Summer Research Initiation Award — May 2020
- Professional level programmer at Samsung Electronics — Jan 2018
- Icon of the month at Samsung R&D Institute Bangladesh Ltd. — Apr 2018
- Received 4 years of OIC scholarship — 2012
- Received 4 years of government scholarship for Higher Secondary Certificate result — 2012

STUDENT MENTORSHIP

Undergraduate Mentees (via University of Central Florida's Software Engineering Project)

- Sparsh Pandey, University of Central Florida — Summer 2025 – Fall 2025
- Terry Achille, University of Central Florida — Fall 2024 – Spring 2025
- Darren Basil, University of Central Florida — Fall 2024 – Spring 2025
- Camilo Alvarez-Velez, University of Central Florida — Fall 2024 – Spring 2025
- James Chen, University of Central Florida — Fall 2024 – Spring 2025
- Patrick Ijieh, University of Central Florida — Fall 2024 – Spring 2025
- Samar Karanch, University of Central Florida — Fall 2024 – Spring 2025

High School Mentees (via George Mason University's Aspiring Scientists Summer Program)

- Alyssa McGowan, Thomas Jefferson High School of Science & Technology — Summer 2023

REFERENCES

Assistant Professor Kevin Moran
Department of Computer Science
University of Central Florida
Room 217A, L3Harris Engineering Center
Orlando, FL 32816, USA
Email: kpmoran@ucf.edu
Phone: (703)-993-6826

Associate Professor Oscar Chaparro
Department of Computer Science
College of William and Mary
McGlothlin-Street Hall 311, 251 Jamestown Rd.
Williamsburg, VA 23185, USA
Email: oscarch@wm.edu
Phone: (757)-221-2144

Professor Andrian Marcus
Department of Computer Science
George Mason University
Room 4452, Nguyen Engineering Building
Fairfax, VA 22030, USA
Email: amarcus7@gmu.edu
Phone: (703)-993-9237

Assistant Professor Mattia Fazzini
Department of Computer Science & Engineering
University of Minnesota
6-207 Keller Hall, 200 Union Street SE
Minneapolis, MN 55455, USA
Email: mfazzini@umn.edu
Phone: (612)-625-4365